Ayesha Raza - AI/ML Engineer

GitHub: github.com/ayesha-ml

Summary
AI/ML Engineer focused on shipping reliable, measurable results for clients.

Skills
Python, PyTorch, scikit-learn, Hugging Face Transformers, LangChain, LangGraph, RAG, Qdrant, FAISS, OpenAI API, LoRA fine-tuning, OpenCV, YOLO object detection, MLflow, NLP, Time-series forecasting, FastAPI (and more)

Selected projects
- Invoice data extraction pipeline for a e-commerce firm (2026): an OCR and layout model that extracts line items from scanned invoices; reduced manual entry by 66% across 800 invoices a month. Tools: Python, OpenCV, PyTorch. [github.com/ayesha-ml/invoice-extractor]
- Real-time shelf monitoring with computer vision for media retail (2026): a detector that flags empty shelves from store cameras; 71% detection mAP at 8 frames per second. Tools: YOLO object detection, OpenCV, PyTorch. [portfolio/shelf-vision]
- Weekly demand forecasting for a logistics distributor (2026): store-level demand forecasts with holiday effects; forecast error (MAPE) reduced to 67%. Tools: Time-series forecasting, Python, pandas. [github.com/ayesha-ml/demand-forecast]
- Internal knowledge assistant for a e-commerce company (2026): a chat assistant that answers staff questions from 800 internal documents; cut average search time from 5 minutes to 6 seconds. Tools: Python, LangChain, RAG. [portfolio/docs-assistant-e-commerce]
- Weekly demand forecasting for a manufacturing distributor (2026): store-level demand forecasts with holiday effects; forecast error (MAPE) reduced to 78%. Tools: Time-series forecasting, Python, pandas. [github.com/ayesha-ml/demand-forecast-manufacturing]
- Review sentiment API for a healthcare brand (2026): an API that scores customer reviews and extracts complaint themes; processes 20000 reviews per day. Tools: NLP, Hugging Face Transformers, FastAPI. [portfolio/sentiment-api-healthcare]
- Evaluation harness for an LLM chatbot (legal) (2025): an offline evaluation set and metrics for answer faithfulness; caught 59% of regressions before release. Tools: Python, RAG, LangChain. [github.com/ayesha-ml/rag-eval]
- Evaluation harness for an LLM chatbot (media) (2025): an offline evaluation set and metrics for answer faithfulness; caught 76% of regressions before release. Tools: Python, RAG, LangChain. [github.com/ayesha-ml/rag-eval-media]
- Support ticket triage with a fine-tuned language model (media) (2024): a fine-tuned classifier that routes tickets to the right queue; routing accuracy of 83% on 800 tickets. Tools: Hugging Face Transformers, LoRA fine-tuning, PyTorch. [github.com/ayesha-ml/ticket-triage]
- Multi-agent document review workflow for a real estate team (2024): agents that review contracts and flag risky clauses with human approval; review time down 90%. Tools: LangGraph, Python, OpenAI API. [github.com/ayesha-ml/agent-workflow]
- Invoice data extraction pipeline for a media firm (2024): an OCR and layout model that extracts line items from scanned invoices; reduced manual entry by 53% across 800 invoices a month. Tools: Python, OpenCV, PyTorch. [github.com/ayesha-ml/invoice-extractor-media]
- Customer churn prediction for a fintech subscription business (2024): a churn model with weekly scoring and retention playbooks; raised retention-campaign precision to 40%. Tools: Python, scikit-learn, pandas. [portfolio/churn-model-fintech]

- Real-time shelf monitoring with computer vision for healthcare retail (2024): a detector that flags empty shelves from store cameras; 40% detection mAP at 4 frames per second. Tools: YOLO object detection, OpenCV, PyTorch. [portfolio/shelf-vision-healthcare]
- Customer churn prediction for a edtech subscription business (2023): a churn model with weekly scoring and retention playbooks; raised retention-campaign precision to 34%. Tools: Python, scikit-learn, pandas. [portfolio/churn-model]
- Semantic product search for a legal marketplace (2023): embedding-based search over 20000 products; search click-through up 44%. Tools: Python, FAISS, Hugging Face Transformers. [portfolio/semantic-search]
- Support ticket triage with a fine-tuned language model (SaaS) (2023): a fine-tuned classifier that routes tickets to the right queue; routing accuracy of 43% on 800 tickets. Tools: Hugging Face Transformers, LoRA fine-tuning, PyTorch. [github.com/ayesha-ml/ticket-triage-saas]
- Multi-agent document review workflow for a logistics team (2023): agents that review contracts and flag risky clauses with human approval; review time down 87%. Tools: LangGraph, Python, OpenAI API. [github.com/ayesha-ml/agent-workflow-logistics]
- Internal knowledge assistant for a edtech company (2022): a chat assistant that answers staff questions from 1200 internal documents; cut average search time from 7 minutes to 16 seconds. Tools: Python, LangChain, RAG. [github.com/ayesha-ml/docs-assistant]
- Review sentiment API for a fintech brand (2022): an API that scores customer reviews and extracts complaint themes; processes 800 reviews per day. Tools: NLP, Hugging Face Transformers, FastAPI. [github.com/ayesha-ml/sentiment-api]
- Semantic product search for a logistics marketplace (2022): embedding-based search over 1200 products; search click-through up 33%. Tools: Python, FAISS, Hugging Face Transformers. [github.com/ayesha-ml/semantic-search-logistics]